	M1 correct	M1 incorrect
M2 correct	both correct	only M2 correct
M2 incorrect	only M1 correct	both incorrect

Table 1: A contingency table representing the distribution of possible outcomes for two models (M1 and M2).

Thus, for McNemar's test, the relevant data generating process for simulations can be specified in terms of the expected difference in accuracy between the models, Δ_{acc}, and P_a, the expected proportion of examples for which the models will have the same outcome (i.e., both correct or both incorrect). From these we can compute the expected proportions of examples on which only one model is correct (i.e., the off-diagonals in Table 1), and estimate power via the algorithm in Figure 2. Figure 3 illustrates how power increases with increased sample size, effect size, and agreement rate.[6]

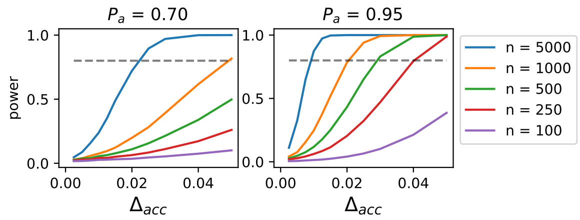

Figure 3: Power for comparing two classifiers on accuracy using paired data depends on the size of the test set (n), the expected agreement (P_a), and the expected difference in accuracy (Δ_{acc}). The dashed line shows 80% power, often taken to be a minimal requirement.

3.2 Estimating parameters

In order to estimate the required parameters (P_a and Δ_{acc}), we consider three options: (1) use results on validation (dev) data; (2) fit a regression based on historical data; (3) use middle-of-the-road assumptions when lacking other information. Using these methods, we can then estimate power or calculate the smallest effect that can be detected with 80% power at $\alpha = 0.05$ (or other thresholds). Both to illustrate this process, and to provide guidance for future work, we demonstrate these approaches below using data from two widely-used datasets for evaluating NLP models: SQuAD 2.0 (Rajpurkar et al., 2016, 2018) and the GLUE benchmark (Wang et al., 2018).

[6]Corresponding plots showing Type-M and Type-S error (Gelman and Carlin, 2014) are in Appendix B. To walk through a numerical example, see Appendix C. For an interactive example, see the accompanying online notebooks.

Using validation results: To the extent that we expect performance on test data to match performance on validation data (i.e., in the absence of domain shift), *paired* performance on validation data (i.e., difference in accuracy and agreement rate) provides one method for estimating power when comparing against a baseline model.

To illustrate this, from the authors of SQuAD 2.0, we obtain the pairwise agreement rates between all models submitted to the leaderboard on both validation and test data. We find a very strong correlation between validation and test for both pairwise accuracy differences (Δ_{acc}) and agreement rates (P_a) ($r = 0.99$ for both, as shown in Figure 9 in Appendix D, with results on validation data included in the accompanying online materials), suggesting we can use paired predictions on validation data for power calculations when we have access to the predictions from both models. Note that this approach assumes that the dev and test data have been drawn from the same distribution, and that dev performance has not been artificially inflated (such as by training on validation data directly).

Using historical data: When one does not have access to the baseline model or an informative prior, one can make use of historical trends. That is, we can try to estimate what a typical improvement will look like, given the current state of the art (SOTA). To illustrate this approach, we collect reported results for both SQuAD 2.0 and GLUE, and fit regressions to estimate Δ_{acc} and P_a. Given these parameters, we can assess the likely power and MDE for a typical model improvement against a given baseline accuracy level.

To fit a regression to predict typical improvements to SOTA, we gather data from GLUE papers and manually label 119 accuracy comparisons and 57 claims of improvement (as denoted by bolding of a result and a claim of SOTA in text) across 14 papers (selected as being at or above the BERT score on the GLUE leaderboard with an accompanying paper). In regressing Δ_{acc} on baseline accuracy and task, we achieve an $R^2 = 0.69$, which is not a perfect fit, but still provides a prior on likely effect size. Similarly, we achieve an $R^2 = 0.67$ when fitting a regression to SOTA improvements on the SQuAD 2.0 leaderboard (selected as being a significant improvement in time-ordered submissions). See Appendix E.2.1 for more details.

To assess power for McNemar's test, we must also fit a regression predicting the expected overlap between the models (P_a). To fit such a regression, from GLUE authors we obtain the model test set predictions on all tasks from a set of 10 high-performing models, which allows us to measure the extent to which their predictions overlap with each other. Using GLUE tasks which measure accuracy, we regress P_a on baseline accuracy and Δ_{acc}, and achieve an R^2 of 0.97.[7] Repeating this for SQuAD 2.0, we get an R^2 of 0.94. See Appendix E.2 for regression coefficients and additional details.

Typical improvements on popular tasks tend to be small (see mean improvements in Table 2). Except for rare transformative work, such as BERT (Devlin et al., 2019), it is generally difficult to do *much* better than a previous SOTA and thus improvements are likely to follow a trend, which is why we are able to use historical data as a guide. In cases where such data is not available or cannot be trusted, other methods are necessary.

No prior: If no informative prior is available and the baseline model or can't be used for comparison on a validation set, then we must fall back on middle of the road assumptions. Lachenbruch (1992) provides a suggested default prior, and we find that MDEs using this method are very similar to those found by using the regression based approach. Appendix E.3 provides more details, and Table 9 in the appendix presents the comparison.

3.3 Assessing power in the literature

Using the regression-based approach of estimating Δ_{acc} and P_a described above, we estimate the MDE for each individual accuracy-based GLUE task in comparison to current SOTA, and report the average effect size of results which claimed improvements. Table 2 summarizes these results, showing for each dataset the size of the test set, the accuracy of the best performing model on each task at the time of writing, the estimated MDE to have 80% power using our regression to predict overlap (P_a), and the average reported difference from their respective baselines.

As can be seen in Table 2, the mean reported effect size ($|\Delta_{acc}|$) is well below the estimated MDE for the three smallest test sets – WNLI, MRPC, and SST-2. Because this mean is based

Dataset	**Size**	**SOTA** (%)	**Est. MDE** (%)	$\lvert\Delta_{acc}\rvert$ (%)
WNLI	147	94.5	+5.26	+1.72
MRPC	1,725	92.0	+1.62	+0.63
SST-2	1,821	97.2	+1.02	+0.57
RTE	3,000	91.7	+1.23	+3.89
QNLI	5,463	97.5	+0.55	+1.31
MNLI-m	9,796	91.6	+0.67	+0.97
MNLI-mm	9,847	91.3	+0.68	+1.29
QQP	390,965	91.0	+0.11	+0.36
SQuAD 2.0	8,862	90.7	+0.56	+2.23†

Table 2: Estimated minimum detectable effect (MDE) using a regression-based estimate of likely agreement with leaderboard SOTA as of May 6th, 2020. $|\Delta_{acc}|$ is the average improvement over baseline per task among surveyed papers that claimed SOTA. For future comparisons, unless the expected improvement is larger than the estimated MDE, an experiment is unlikely to be adequately powered, and researchers should instead choose a different (larger) dataset. Note that this likely applies to the vast majority of experiments on WNLI, MRPC, and SST-2, based on recent trends. † indicates that the SQuAD 2.0 average was based on leaderboard improvements, which weren't necessarily reported in a publication. See Appendix E for full table and details.

on models comparing to even weaker baselines, we would expect most future improvements to be even smaller. Thus, most future experiments involving these three datasets *will not have adequate power* to test for improvements over the current SOTA in the way that they are routinely used. Moreover, alternative analyses give *even more pessimistic* estimates of likely improvements relative to MDE, as described in Appendix E.4. If an experiment does show significant improvement on a dataset such as MRPC, the potential for Type-M error should make us skeptical that this improvement will generalize to new data from the same domain.

While the above results are informative about future experiments, we would also ideally like to know about the power of past experiments. Most of the papers from which we collected results did not report a significance test on the test set. Here we estimate the expected power and predicted result of such a test using leave-one-out regressions, where we make a prediction for each reported improvement using all other reported model comparisons. This procedure reveals that **only 46% would have predicted adequate power** (using estimates for expected improvement and agreement), and **approximately 51% would have been significant** (based on estimated agreement and *reported* improvement). Approximately 80% of experiments with at least 80% power would also have been

[7] WNLI (Levesque et al., 2012), MRPC (Dolan and Brockett, 2005), SST-2 (Socher et al., 2013), RTE (Dagan et al., 2005; Bar-Haim et al., 2006; Giampiccolo et al., 2007; Bentivogli et al., 2009), QNLI (Rajpurkar et al., 2016) MNLI (Williams et al., 2018), and QQP (Iyer et al., 2017). For consideration of other metrics, see Appendix F.

found to be significant (37% of all comparisons).

In part because performance on many of these tasks is now so good, a large expected improvement is required in order for a new experiment to have 80% power, suggesting that larger test set sizes may be necessary to continue making well-powered claims of SOTA improvement on individual tasks. For any comparisons which are likely to be underpowered, we should refrain from placing much emphasis on obtaining small improvements over the previously reported best model. In extreme cases, such as MRPC and SST-2, it is worth considering whether it is time to retire these datasets as the basis for model comparison.[8]

4 Machine Translation

To show how our approach to power analysis can be applied to a more difficult setting, we consider automated evaluation of machine translation using BLEU scores (Papineni et al., 2002). As with accuracy, we would like to know what scale of improvements can be detected with reasonable power on typical test sets. This setting is more complicated because (1) BLEU is a corpus-level metric, rather than being averaged across instances, and (2) typical models are trained on vast amounts of parallel data, with little data available that has not been used in training, making it difficult to estimate variation in performance.

Significance testing for BLEU: To test for a significant difference between two MT models we use the randomization test, as recommended in Dror et al. (2018): given the paired output translations from both models, swap the outputs for a random subset of test examples and compute the resulting difference in BLEU. Repeating this thousands of times gives us a null distribution, which can be used to test the observed difference between models.

Generative process for simulations: If large amounts of untouched evaluation data were available, we could approach power analysis by simply evaluating BLEU score on many random subsets of n sentences, and computing the mean and variance of each system. Unfortunately, because MT depends on parallel text (most of which is used in training), evaluation data tends to be scarce. Instead, we introduce a generative process that can produce the necessary inputs for power analysis.

For intuition, note that if we swap the i^{th} pair of model outputs (as is done in the randomization test), leaving rest as they are, we change the difference in BLEU between models by a specific amount, δ_i, which we call the effect of making that swap. While these individual effects are not independent of each other due to the corpus-level nature of the metric, in practice, the sum of individual effects closely approximates the net effect of swapping entire subsets (see Figure 15 in Appendix G).

Based on analyzing several models and datasets, we find the typical distribution of these individual effects can be approximated using a mixture of a Delta distribution at zero, and a Laplace distribution (see Appendix G for details). Concretely, if we assume Δ_B is the expected difference in BLEU between two models on a dataset of n examples, and P_0 is the expected proportion of examples for which $\delta_i = 0$, we can simulate a dataset $\{\delta_i\}_{i=1}^{n}$ of n individual effects using the following process: with probability P_0, $\delta_i = 0$. With probability $1 - P_0$, $\delta_i \sim \text{Laplace}(\mu, b)$, where $\mu = \frac{-2 \cdot \Delta_B}{n(1-P_0)}$, $b = b_0/n$, and b_0 is a user-specified parameter that controls the variance, independent of the sample size. By construction, $\mathbb{E}[\sum_{i=1}^{n} \delta_i] = -2 \cdot \Delta_B$.[9]

Given this generative process, we can then estimate power using the Algorithm in Figure 2. On each iteration, draw a simulated dataset from the generative process, compute the observed difference between models as $\hat{\Delta}_B = -\frac{1}{2} \sum_{i=1}^{n} \delta_i$, and test if this is significantly different from zero using a modified randomization test, in which we assume that the net effect of swapping a subset of instances is simply the sum of the δ_i's in the subset. (Please see online materials for an interactive example).

Empirical estimates: In order to estimate reasonable values for the required parameters, we use several pretrained models from the FAIRSEQ library (Ott et al., 2019) for the WMT English-German translation task. We evaluate these models on the shared task test sets from 2016-2019 and compute BLEU scores using SACREBLEU (Post, 2018). Fitting a Delta-Laplace mixture to the effects of swapping individual output pairs, we estimate values for $\hat{P}_0$ and $\hat{b}_0$, reported in Table 3. (See also Figure 16 in Appendix G; code for computing estimates is provided in the online materials).

[8]It is also worth exploring power with respect to claims of improvement on multiple tasks with a single model (Demšar, 2006), rather than each task individually. We leave consideration of this as an interesting direction for future work.

[9]Note that swapping all n examples would reverse the model scores, equivalent to a net effect of $-2 \cdot \Delta_B$.